

PHILIP, SD, USA

WMO: 726516

Lat: 44.051N

Lon: 101.601W

Elev: 2206

StdP: 13.56

Time Zone: -7.00 (NAM)

Period: 98-19

WBAN: 24024

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.4	(c) -3.8	(d) -15.6	(e) 2.5	(f) -8.7	(g) -10.3	(h) 3.4	(i) -3.3	(j) 34.6	(k) 27.8	(l) 31.8	(m) 30.1	(n) 6.4	(o) 270	(p) 0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 28.5	(c) 100.0	(d) 68.9	(e) 96.4	(f) 68.7	(g) 92.2	(h) 67.8	(i) 73.9	(j) 88.7	(k) 72.2	(l) 87.7	(m) 70.6	(n) 86.2	(o) 12.4	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.9	119.6	80.4	67.7	110.4	78.5	65.8	103.3	77.3	38.9	88.4	37.4	87.8	35.9	86.1	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.7	26.1	23.0	-20.1	105.6	6.1	4.5	-24.5	108.9	-28.1	111.5	-31.6	114.0	-36.0	117.3
			-20.2	76.7	5.4	2.0	-24.1	78.2	-27.2	79.4	-30.3	80.5	-34.2	82.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	23.4	24.6	36.0	46.3	56.4	67.5	76.2	73.4	63.9	48.5	35.8	23.9
	DBStd	21.34	13.07	13.94	12.58	10.10	8.50	7.54	6.62	6.68	9.06	9.63	11.16	12.61
	HDD50	3595	825	713	452	182	34	0	0	0	6	140	435	808
	HDD65	7033	1289	1132	898	564	288	57	4	11	127	515	875	1273
	CDD50	2913	0	2	19	71	232	524	812	725	423	95	10	0
	CDD65	876	0	0	0	3	21	131	351	271	94	5	0	0
	CDH74	12390	0	0	27	132	451	1867	4640	3511	1574	181	7	0
	CDH80	6217	0	0	4	33	147	834	2563	1836	742	57	1	0
Wind	WSAvg	9.9	9.8	9.8	10.6	11.6	10.9	9.6	8.9	8.9	9.3	10.0	9.5	9.5
Precipitation	PrecAvg	15.4	0.4	0.4	0.9	1.7	2.8	3.0	1.7	1.6	1.1	1.3	0.5	0.4
	PrecMax	22.3	0.9	1.3	2.6	3.9	6.2	5.6	4.4	4.3	4.0	4.1	1.7	1.2
	PrecMin	6.9	0.0	0.0	0.0	0.4	0.3	0.3	0.0	0.3	0.0	0.0	0.0	0.0
	PrecStd	3.8	0.2	0.3	0.6	0.9	1.8	1.5	1.3	0.9	1.1	1.2	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.4	67.5	78.0	84.5	90.5	101.1	107.1	103.0	99.5	86.8	73.5	60.5
		MCWB	44.3	47.8	53.4	57.2	66.1	69.3	69.6	68.0	66.3	59.5	51.4	44.9
	2%	DB	51.7	57.4	70.4	78.2	84.1	93.7	101.0	98.9	93.4	79.2	65.8	52.2
		MCWB	40.9	43.2	50.6	56.3	63.9	69.1	69.8	68.5	64.8	56.7	48.7	40.9
	5%	DB	45.6	50.3	63.5	71.7	79.0	89.1	97.0	94.4	88.5	72.6	60.3	46.2
		MCWB	37.2	39.9	47.5	53.4	61.5	68.2	69.9	67.9	64.0	54.2	46.0	37.7
	10%	DB	41.3	44.5	56.8	65.5	74.0	84.2	92.7	90.3	83.3	66.5	54.0	41.2
		MCWB	35.2	36.8	44.4	50.3	59.1	66.3	69.0	67.3	62.3	52.0	42.9	34.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.0	48.4	55.3	59.9	69.9	75.7	76.9	74.3	70.7	62.0	53.2	45.5
		MCDB	58.6	65.2	72.1	78.7	83.7	86.8	91.0	90.7	88.0	81.2	71.2	58.4
	2%	WB	41.0	44.2	51.5	57.0	66.1	72.3	74.7	72.5	68.4	58.7	49.3	41.2
		MCDB	50.8	56.5	67.8	74.6	79.6	86.7	89.4	89.0	85.5	74.9	64.8	51.1
	5%	WB	38.0	40.3	48.4	54.6	63.2	70.1	72.9	70.9	66.2	55.6	46.3	38.0
		MCDB	44.8	50.1	63.0	69.9	76.4	85.3	87.9	86.8	83.0	69.3	59.1	46.1
	10%	WB	35.3	36.9	45.0	51.4	60.3	68.1	71.3	69.3	64.2	53.0	43.3	34.9
		MCDB	40.7	43.7	56.7	63.7	71.4	82.3	87.1	84.2	79.8	65.2	53.3	40.8
Mean Daily Temperature Range	5% DB	MDBR	22.6	23.7	25.4	25.8	25.4	26.6	28.5	29.2	29.8	26.9	25.8	22.4
		MCDBR	28.6	33.5	38.2	38.1	33.9	34.2	34.1	35.7	38.3	38.2	36.0	29.6
		MCWBR	19.0	20.7	21.4	19.4	16.7	13.7	10.7	12.0	14.8	18.7	20.9	19.9
	5% WB	MCDBR	27.3	31.8	36.5	34.9	30.8	29.8	28.6	29.7	32.1	34.2	34.8	28.8
		MCWBR	18.7	19.9	21.0	18.3	16.3	14.2	11.8	11.9	13.5	17.5	20.6	19.7
Clear-Sky Solar Irradiance	taub		0.260	0.266	0.288	0.329	0.351	0.368	0.386	0.379	0.342	0.296	0.268	0.257
	taud		2.555	2.539	2.529	2.402	2.412	2.393	2.377	2.379	2.460	2.564	2.554	2.542
	Ebn at Noon		284	300	305	297	292	286	279	277	281	284	277	273
	Edh at Noon		20	25	29	35	36	37	37	36	30	24	20	19
All-Sky Solar Radiation	RadAvg		601	912	1266	1588	1905	2176	2219	1886	1495	984	668	504
	RadStd		42	66	98	102	181	149	86	112	98	108	56	36

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	-2.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page